PRODUCT DATASHEET · SKU PIA24023P

8-Channel Wireless Wearable BCI

A coin-sized, battery-powered brain-computer interface. A Texas Instruments ADS1299 reads 8 channels of EEG, EMG, ECG or EOG at 24 bits, and an STM32WB55 streams them over Bluetooth Low Energy to a browser, phone or laptop. No Raspberry Pi, no cables to the computer.



8	24-bit	250 SPS	BLE 5	Ø 50
Channels + REF + BIAS	TI ADS1299 front end	All channels together	STM32WB55 wireless MCU	mm · 3-board stack

01 OVERVIEW

Key features

- **Wireless and wearable**: Ø 50 mm stack on a small LiPo, mounts on a cap or 3D-printed headset.
- **Research-grade front end**: TI ADS1299, 8 simultaneous 24-bit channels, internal noise below 1 µV.
- **Bluetooth Low Energy**: no pairing, connects straight from PiEEG Cloud, Android, Windows, macOS or Linux.
- **Open software**: PiEEG Server (LSL, OSC, WebSocket, CSV), Python BLE SDK, JavaScript SDK, Android app.
- **Updatable firmware**: SWD header for ST-Link; I²C expansion header for extra sensors.
- **Open data**: dry-electrode alpha dataset and example projects on GitHub.

Typical applications

- Mobile and wearable BCI prototypes
- Neurofeedback and alpha training
- EEG / EMG / EOG teaching labs
- Games, VR and OSC control
- Field recordings away from mains
- Machine learning on biosignals

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	8 biosignal inputs
Extra inputs	Reference (common to all channels) and bias drive
Analog front end	Texas Instruments ADS1299
Resolution	24-bit delta-sigma, all channels sampled at the same moment
Sample rate	250 SPS
Internal noise	Below 1 µV (IronBCI paper); 1 LSB ≈ 0.268 µV
Signals	EEG, EMG, ECG, EOG
WIRELESS	
Microcontroller	ST STM32WB55 (Arm Cortex-M4 application core + Cortex-M0+ radio core)
Radio	Bluetooth Low Energy 5, 2.4 GHz, on-board chip antenna
Advertised name	`EAREEG` (some units `IronBCI`)
Pairing	Not needed — connect and subscribe to notifications

For research and education only. Not a medical device. Specifications subject to change.

Data	24 bytes per sample (8 × 3-byte channels), 250 samples per second
Range	A few metres, line of sight
ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connector	1 × 10 pin header, 2.54 mm: E1–E8, Biasout, Ref
POWER & SAFETY	
Battery	Single-cell LiPo, 200 mAh recommended, 2.0 mm JST, underside (not included)
Charging	Micro USB, 5 V, on the power board; LED 1 charging, LED 2 charged
Run time	≈ 1–1.5 h per charge with 200 mAh (estimate; larger cells run longer)
Switches	Switch 1: battery on/off · Switch 2: data flow on/off
Isolation	Battery only while worn. Never on a person while charging
Classification	Research and education tool. Not a medical device
MECHANICAL	
Construction	Three round Ø 50.0 mm boards: power, ADC, BLE (bottom to top)
Mounting	4 × Ø 3.0 mm holes, 29.86 × 33.04 mm pattern; M2 spacers between boards
Shipping weight	100 g (packed)
SOFTWARE	
PiEEG Cloud	cloud.pieeg.com → PiEEG Devices → IronBCI (Chrome / Edge)
PiEEG Server	`pip install "pieeg-server[ironbci]"` · `--device ironbci8`
SDKs	Python (bleak) BLE scripts · JavaScript pieeg.js (default board) · Android app (MIT)

03 BLOCK DIAGRAM

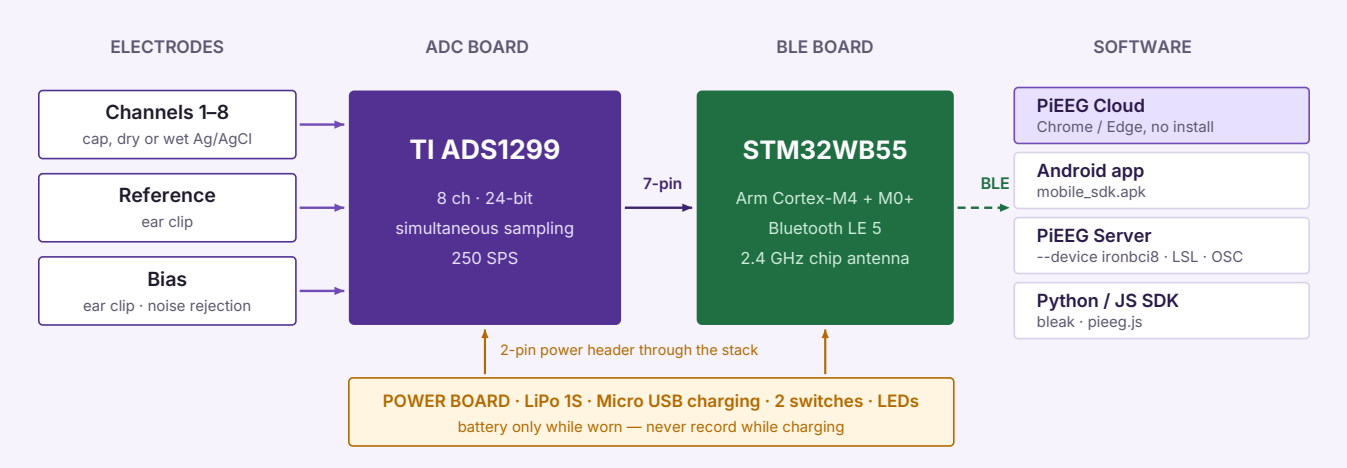


Figure 1 — Signal path. The ADS1299 digitises all channels together; the STM32WB55 packs each sample and sends it over BLE.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms: delta, theta, alpha, beta, gamma	Close the eyes and watch alpha (8–13 Hz) rise on Pz
EOG	Eye movements and blinks	Blink or look left/right — large slow waves on F7/F8
EMG	Muscle activity	Clench the jaw or a fist
ECG	Heart activity	Electrodes on the chest or wrists

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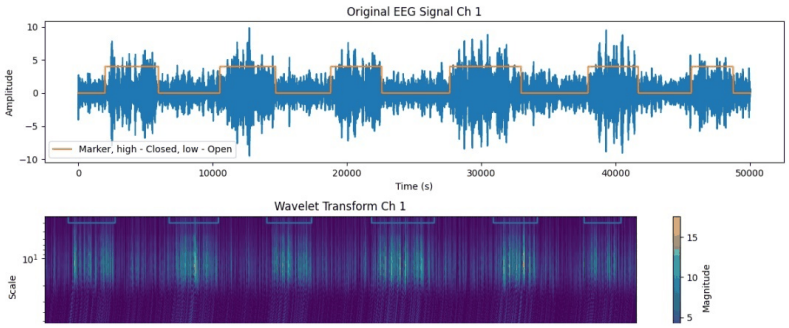


Figure 2 — Alpha test with dry electrodes: eyes open / closed (top), wavelet transform (bottom).

05 PIEEG FAMILY

	IronBCI-8	PiEEG-8	IronBCI-32
Channels	**8**	8	32
Host link	**Bluetooth LE**	Raspberry Pi (SPI)	Laptop (USB)
Converter	**1 × ADS1299**	1 × ADS1299	4 × AD7771
Power	**LiPo, wearable**	5 V power bank	Laptop battery
List price	**USD 420**	USD 350	USD 480

06 ORDERING INFORMATION

Product	IronBCI-8 — 8-channel wireless wearable BCI
SKU	PIA24023P
Price	USD 420 (list)
Purchase	Elecrow (official manufacturing partner) — elecrow.com/pieeg.html · direct store on pieeg.com coming soon

Included	
1 × IronBCI-8	Power, ADC and BLE boards, assembled
Quick-start manual	Printed guide

Not included — required	
LiPo battery	1S, 200 mAh, 2.0 mm JST
Micro USB cable	For charging only
Electrodes and cap	8 Ag/AgCl + 2 ear clips
BLE receiver	PC, phone or Chrome / Edge

WARNING Never wear while charging.

IronBCI-8 must run from its LiPo battery with the USB cable unplugged whenever electrodes are on a person. IronBCI-8 is not a medical device.

07 RELATED DOCUMENTS

IB8-MD-002 Mechanical & Assembly Drawing	**IB8-PCB-003** PCB & Manufacturing Specification	**IB8-PIN-004** Connector, Electrode & Battery Guide
IB8-QSG-005 Quick Start & Safety Guide	**IB8-SW-006** Bluetooth Interface & Software Guide	